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Tanahashi et al.

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(54) **SCROLL COMPRESSOR**

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CPC ***F04C 29/02*** (2013.01); ***F04C 18/0215*** (2013.01); ***F04C 27/005*** (2013.01); ***F04C 29/028*** (2013.01); *F04C 23/008* (2013.01)

(58) **Field of Classification Search**
CPC F04C 18/0215; F04C 23/008; F04C 23/02; F04C 27/005; F04C 29/02; F04C 29/023; F04C 29/026; F04C 29/028
See application file for complete search history.

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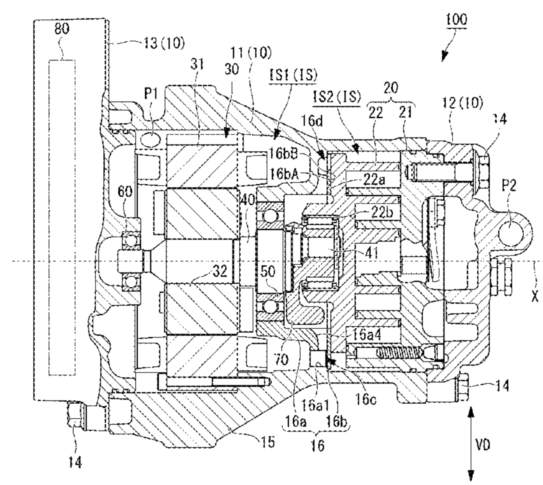
(57) **ABSTRACT**

Provided is a scroll compressor including: a motor configured to revolve an orbiting scroll with respect to a fixed scroll; a rotary shaft rotated about an axis by the motor; a bearing part configured to support the rotary shaft; and a housing. The housing has a partition wall part-partitioning the inner space into a first space and a second space, the motor being arranged in the first space, and the bearing part and a compressing part being arranged in the second space. The partition wall part has a first communication passage
(Continued)



and a second communication passage, the first communication passage being configured to guide a mixed refrigerant containing a lubricating oil and a refrigerant gas from the first space to a region of the second space, the bearing part being arranged in the region of the second space, and the second communication passage being configured to guide the refrigerant gas, which is contained in the mixed refrigerant guided to the region of the second space, to the compressing part.

8 Claims, 6 Drawing Sheets

(51) **Int. Cl.**
F04C 29/02 (2006.01)
F04C 23/00 (2006.01)

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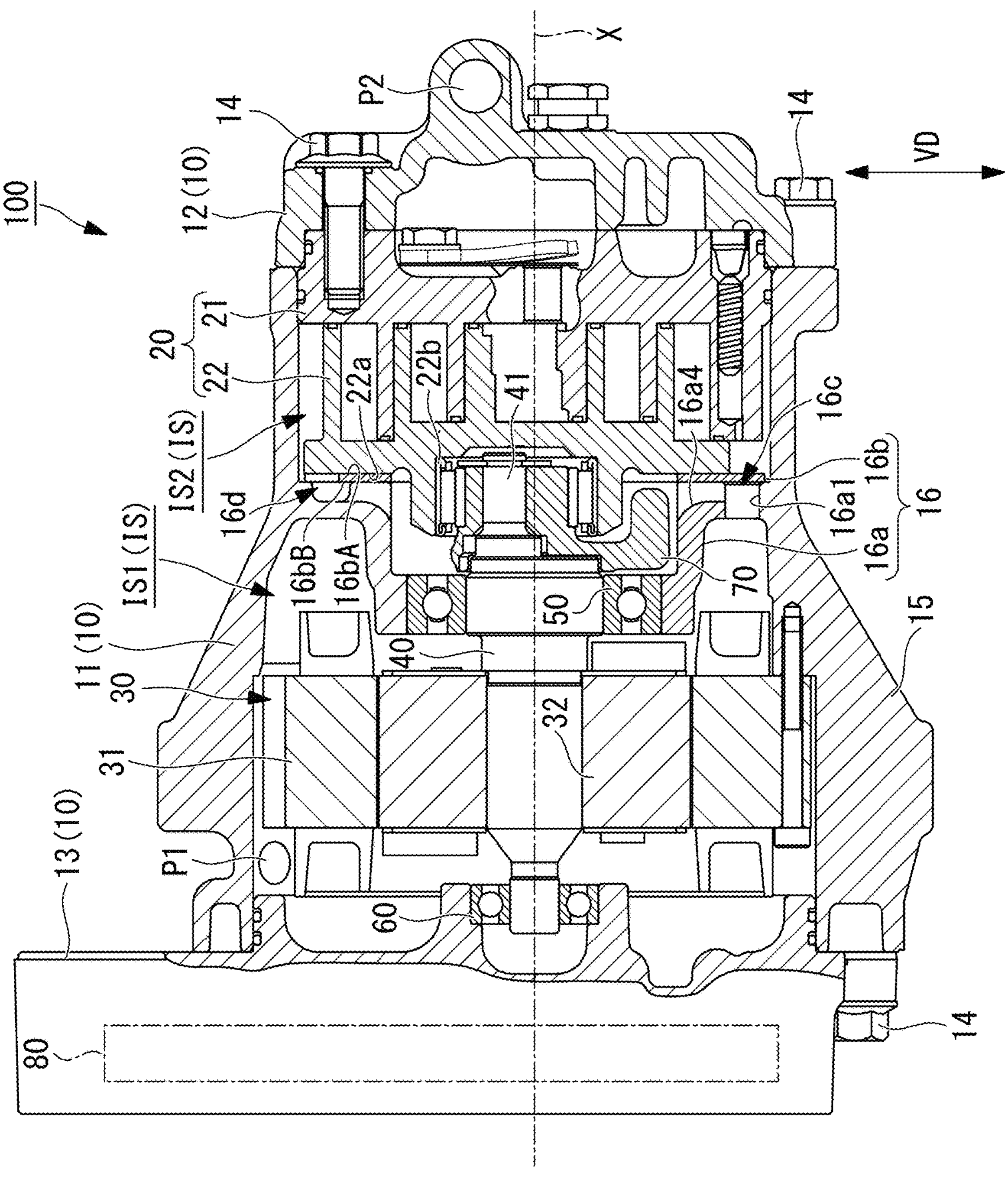

100
X
P2
14
12 (10)
20
21
22
22a
22b
41
16a4
16c
16b
16
16a1
16a
70
50
40
16bB
16bA
16d
IS2 (IS)
IS1 (IS)
11 (10)
30
31
32
15
VD
13 (10)
P1
60
80

FIG. 1

FIG. 2

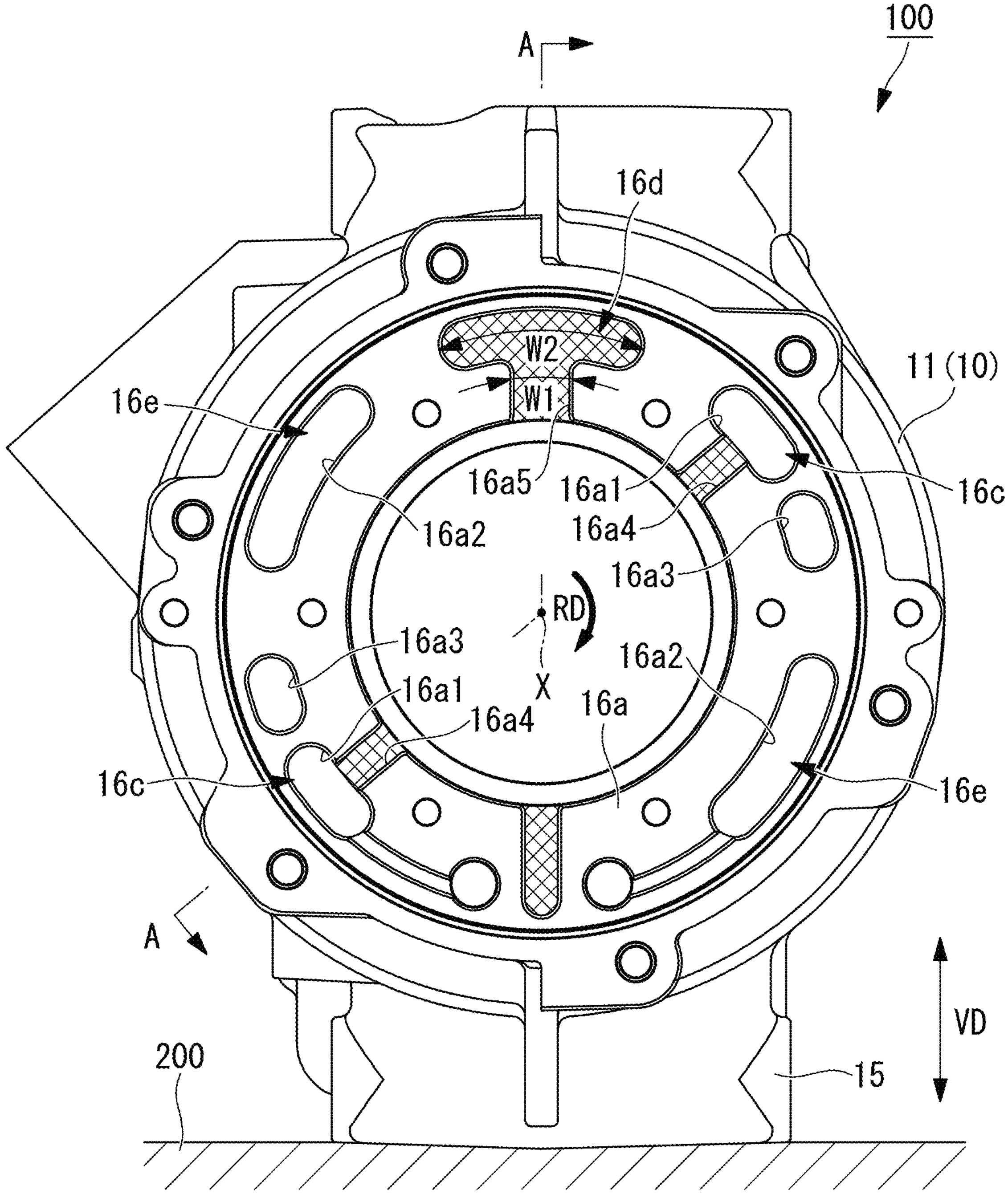

100
A
16d
W2
W1
11(10)
16e
16a5
16a1
16c
16a2
16a4
16a3
RD
16a3
16a2
16a1
X
16a4
16a
16c
16e
A
VD
200
15

FIG. 3

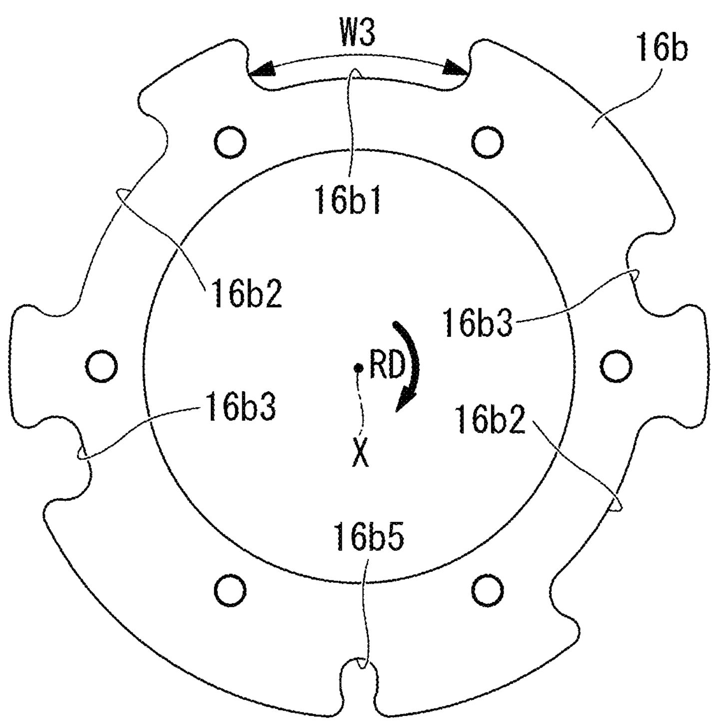


FIG. 4

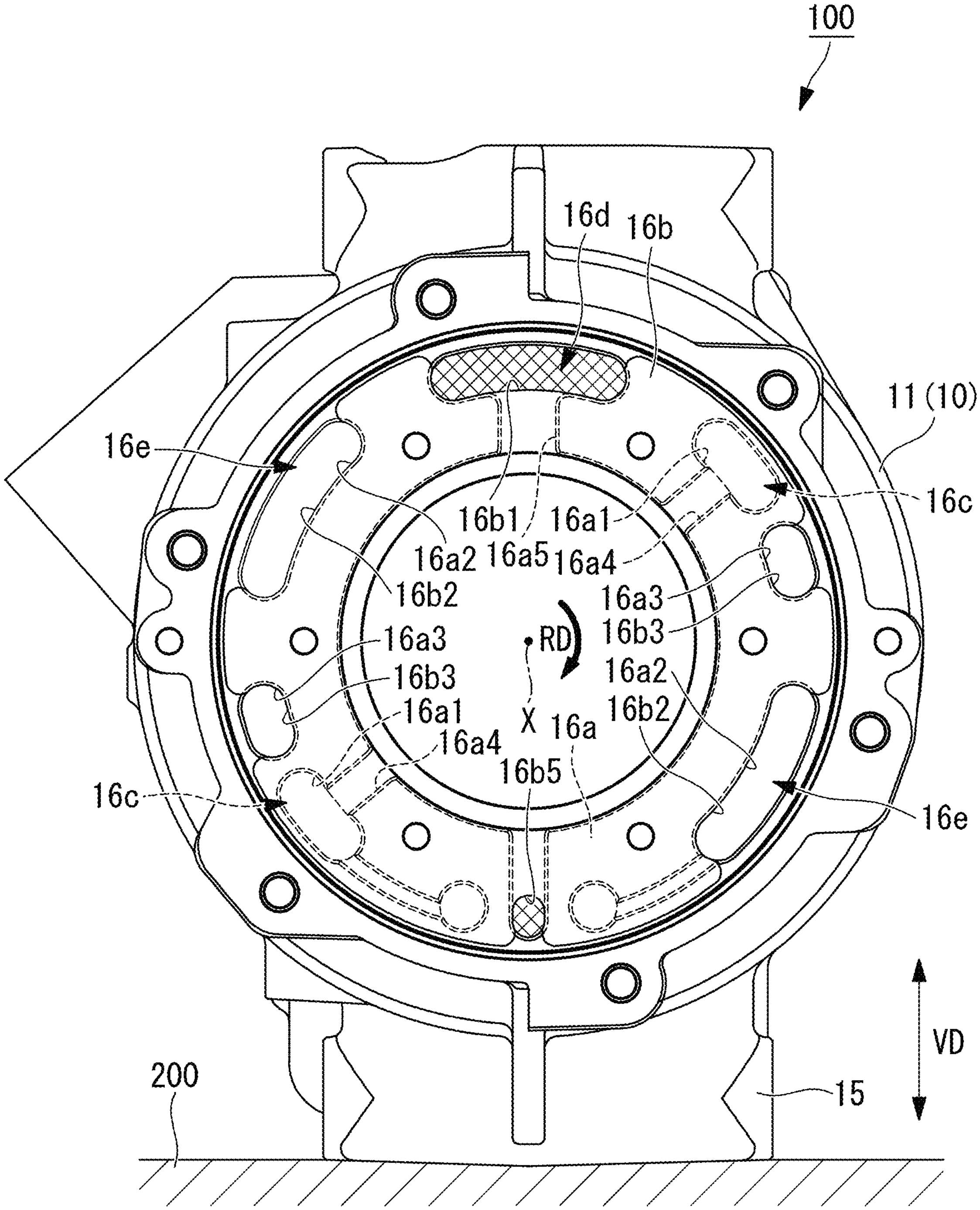

100
16d
16b
11 (10)
16e
16c
16b1
16a1
16a2
16a5
16a4
16a3
16b2
16b3
RD
16a3
16b3
16a2
16a1
16b2
X
16a
16a4
16b5
16c
16e
VD
200
15

FIG. 5

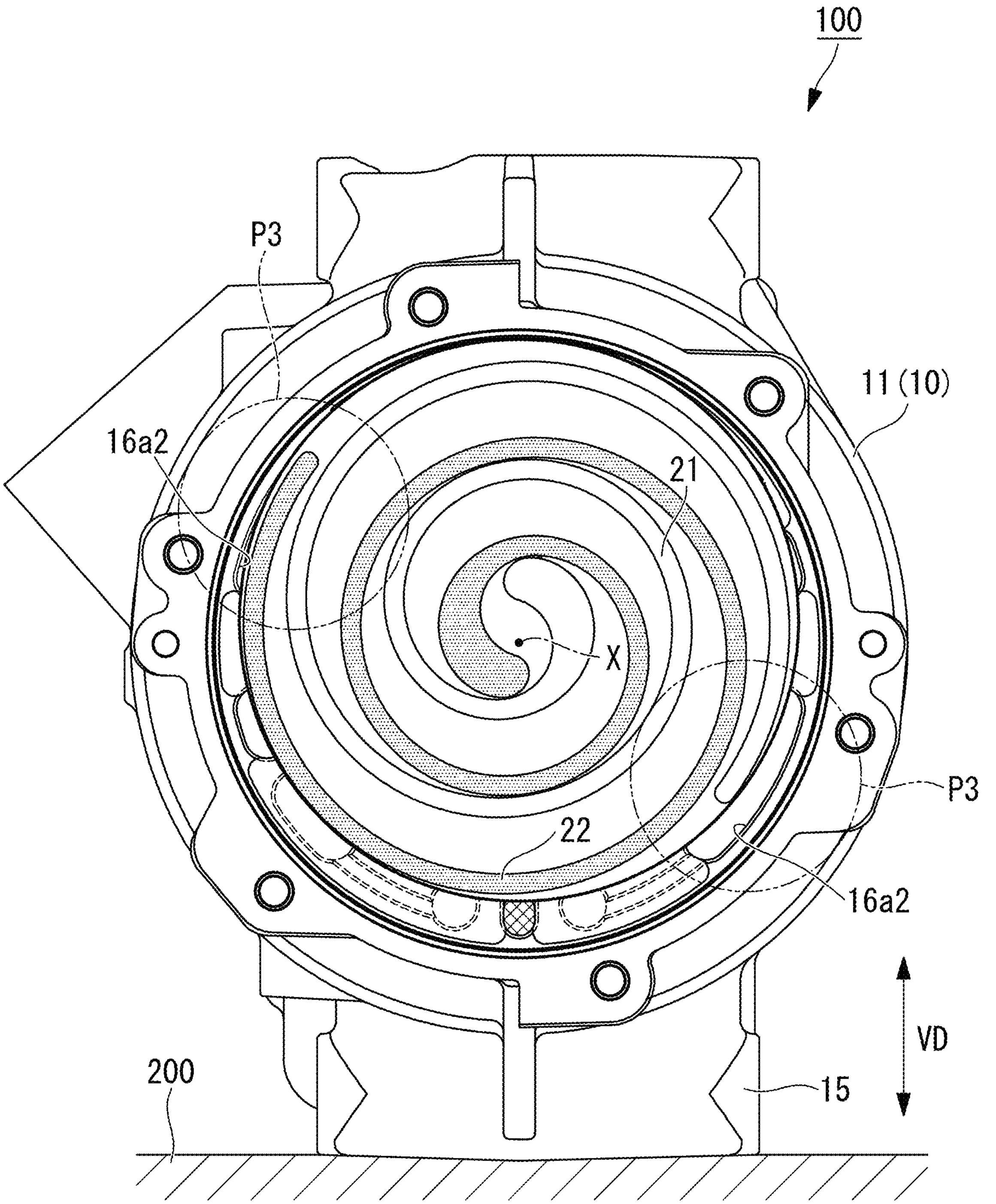
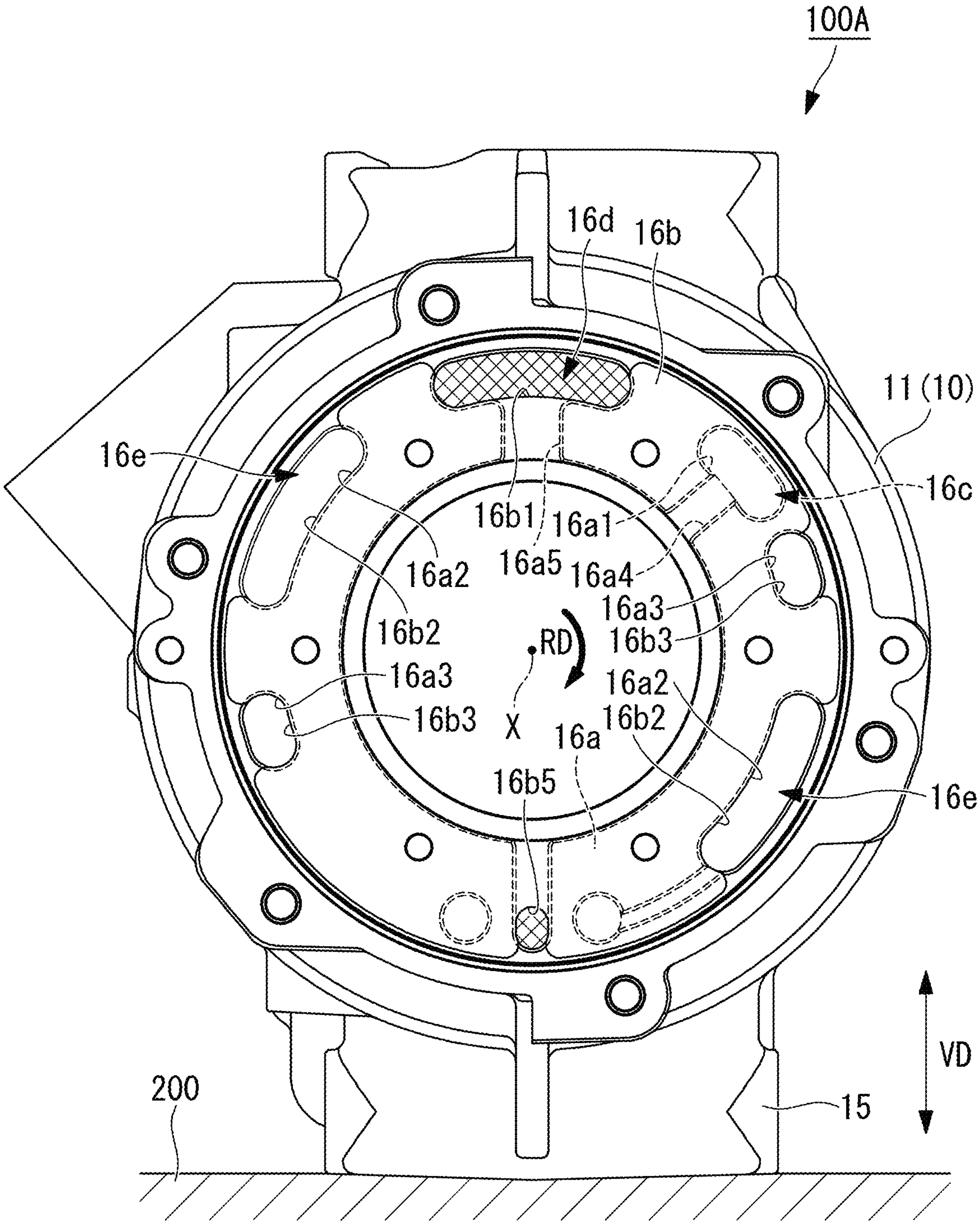


FIG. 6


100A
16d
16b
11 (10)
16e
16c
16b1
16a1
16a5
16a4
16a2
16a3
16b2
16b3
RD
16a3
16a2
16b3
16b2
X
16a
16b5
16e
VD
200
15

SCROLL COMPRESSOR

TECHNICAL FIELD

The present disclosure relates to a scroll compressor.

BACKGROUND ART

Scroll compressors having a fixed scroll and an orbiting scroll engaged with the fixed scroll are conventionally known (for example, see Patent Literature 1). In Patent Literature 1, a housing on the compressor side and a housing on the electric motor side are fixed interposing a partition wall member therebetween, and thereby the housing on the compressor side and the housing on the electric motor side are separated from each other. A plurality of air inlets to communicate chambers on both sides with each other are provided in the partition wall member.

After cooling the electric motor while flowing through the housing on the electric motor side, a refrigerant gas flows into the housing on the compressor side via the air inlets of the partition wall member. The refrigerant gas that has flown into the housing on the compressor side is drawn in a variable volume chamber between the fixed scroll and the orbiting scroll, compressed into a high pressure, and discharged from the end of the housing on the compressor side. In the scroll compressor of Patent Literature 1, the refrigerant gas contains oil, and the oil turns to a mist state, which floats inside the housing and lubricates each portion inside the housing.

CITATION LIST

Patent Literature

PTL 1

Japanese Patent Application Laid-Open No. 2011-174453

SUMMARY OF INVENTION

Technical Problem

In Patent Literature 1, however, the refrigerant gas guided to the plurality of air inlets formed in the partition wall member is drawn in the variable volume chamber between the fixed scroll and the orbiting scroll without passing through the region in which a main bearing for supporting a main shaft connected to the rotary shaft of the motor, a drive bearing installed to a rear face of the orbiting scroll, or the like are arranged. Thus, the oil contained in the refrigerant gas is unable to be efficiently supplied to lubricate the main bearing or the drive bearing.

The present disclosure has been made in view of such circumstances and intends to provide a scroll compressor that can reliably guide a lubricating oil contained in a mixed refrigerant to a bearing part and reliably guide a refrigerant gas contained in the mixed refrigerant to a compressing part.

Solution to Problem

A scroll compressor according to one aspect of the present disclosure includes: a compressing part having a fixed scroll and an orbiting scroll engaged with the fixed scroll; a motor configured to revolve the orbiting scroll with respect to the fixed scroll; a rotary shaft configured to be rotated about an axis by the motor and installed to the orbiting scroll via an eccentric shaft arranged eccentrically from the axis; a bearing part configured to support the rotary shaft; and a housing formed in a cylindrical shape along the axis and having an inner space accommodating the motor and the compressing part. The housing has a partition wall part partitioning the inner space into a first space and a second space, the motor being arranged in the first space, and the bearing part and the compressing part being arranged in the second space. The partition wall part has a first communication passage and a second communication passage, the first communication passage being configured to guide a mixed refrigerant containing a lubricating oil and a refrigerant gas from the first space to a region of the second space, the bearing part being arranged in the region of the second space, and the second communication passage being configured to guide the refrigerant gas, which is contained in the mixed refrigerant guided to the region of the second space, to the compressing part.

Advantageous Effects of Invention

According to the present disclosure, it is possible to provide a scroll compressor that can reliably guide a lubricating oil contained in a mixed refrigerant to a bearing part and reliably guide a refrigerant gas contained in the mixed refrigerant to a compressing part.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a longitudinal sectional view illustrating a general configuration of a scroll compressor according to one embodiment of the present disclosure.

FIG. **2** is a diagram of a housing of the scroll compressor illustrated in FIG. **1** when viewed from the compressing part side.

FIG. **3** is a plan view of a thrust plate illustrated in FIG. **1**.

FIG. **4** is a diagram illustrating a state where the thrust plate is installed to the housing illustrated in FIG. **2**.

FIG. **5** is a diagram illustrating a state where the compressing part is installed to the housing illustrated in FIG. **4**.

FIG. **6** is a diagram of a housing of a scroll compressor of a modified example when viewed from the compressing part side.

DESCRIPTION OF EMBODIMENTS

A scroll compressor according to one embodiment of the present disclosure will be described with reference to FIG. **1** to FIG. **5**. FIG. **1** is a longitudinal sectional view illustrating a general configuration of the scroll compressor according to the present embodiment. A scroll compressor **100** of the present embodiment is used for an on-vehicle air conditioner, for example. FIG. **2** is a diagram of a housing of the scroll compressor illustrated in FIG. **1** when viewed from the compressing part side. FIG. **1** is a sectional view of the scroll compressor **100** illustrated in FIG. **2** taken along the line indicated by arrows A-A. In FIG. **2**, the rotation direction RD represents a rotation direction of a rotary shaft **40**.

As illustrated in FIG. **1**, the scroll compressor **100** has a housing **10**, a compressing part **20**, a motor **30**, the rotary shaft **40**, a bearing part **50**, a bearing part **60**, a balance weight **70**, and an inverter **80**.

The housing **10** is an outer shell of the scroll compressor **100** and is formed of an aluminum alloy. The housing **10** has a first housing **11**, a second housing **12**, and a third housing **13**. The first housing **11**, the second housing **12**, and the third housing **13** are configured to be integrally fastened with bolts **14**. As illustrated in FIG. **2**, the housing **10** is fixed to a casing **200** via a leg **15** in a state where the axis X is arranged in the horizontal direction.

As illustrated in FIG. **1**, an intake port P**1** is provided on the upper side in the vertical direction (gravity direction) VD in the housing **10**. The refrigerant supplied outside is introduced to an inner space IS of the housing **10** from the intake port P**1**. The refrigerant introduced to the housing **10** passes through the motor **30** along the axis X and is guided toward the compressing part **20**. The refrigerant taken into from the intake port P**1** is a mixed refrigerant containing a lubricating oil and the refrigerant gas.

The first housing **11** has the inner space IS formed in a substantially circular cylindrical shape along the axis X and accommodating the compressing part **20** and the motor **30**. The second housing **12** seals one end of the first housing **11** along the axis X and is provided with a discharge port (not illustrated) for the refrigerant gas compressed by the compressing part **20**. The third housing **13** seals the other end of the first housing **11** along the axis X and is provided with a space accommodating the inverter **80** inside thereof.

The housing **10** has a partition wall part **16** partitioning the inner space IS into a first space IS**1** in which the motor **30** is arranged and a second space IS**2** in which the bearing part **50** and the compressing part **20** are arranged. The partition wall part **16** has a bearing support part **16***a* formed integrally with the first housing **11** and a thrust plate **16***b*. The details of the partition wall part **16** will be described later.

The compressing part **20** is a device arranged inside the first housing **11** and rotated about the axis X to compress the refrigerant gas. The compressing part **20** has a fixed scroll **21** fixed in the second housing **12** and an orbiting scroll **22** engaged with the fixed scroll **21**. The compressing part **20** compresses the refrigerant gas by revolving the orbiting scroll **22** with respect to the fixed scroll **21** by driving force of the motor **30**.

The motor **30** is a device that revolves the orbiting scroll **22** of the compressing part **20** about the axis X with respect to the fixed scroll **21**. The motor **30** has a stator **31** and a rotor **32**. The rotor **32** is connected to the rotary shaft **40**.

The rotary shaft **40** is a shaft-like member rotated about the axis X by the motor **30**. One end of the rotary shaft **40** is supported by the bearing part **60** fixed to the third housing **13**. The other end of the rotary shaft **40** is supported by the bearing part **50** fixed to the bearing support part **16***a*. An eccentric shaft **41** arranged eccentrically with respect to the axis X is provided at the end on the compressing part **20** side of the rotary shaft **40**.

The eccentric shaft **41** is rotatably installed to a bearing part **22***b* fixed to the rear face of the orbiting scroll **22** via the balance weight **70**. The rotary shaft **40** is installed to the orbiting scroll **22** via the eccentric shaft **41** in such a way.

The bearing part **50** is a member that is press-fitted into the bearing support part **16***a* and supports the rotary shaft **40** rotatably about the axis X.

The bearing part **60** is a member that is press-fitted into the third housing **13** and supports the rotary shaft **40** rotatably about the axis X.

The balance weight **70** is a member fixed to the eccentric shaft **41** of the rotary shaft **40** and rotated about the axis X. The balance weight **70** cancels vibrations due to revolution motion of the orbiting scroll **22**.

The inverter **80** is a device that generates a drive voltage for driving the motor **30** and controls a rotation rate of the motor **30**.

Next, the details of the partition wall part **16** will be described.

The bearing support part **16***a* is a member that partitions the inner space IS into the first space IS**1** and the second space IS**2** and into which the bearing part **50** is press-fitted. As illustrated in FIG. **2**, two through-holes **16***a***1**, two through-holes **16***a***2**, and two through-holes **16***a***3** are formed in a face of the bearing support part **16***a* on the thrust plate **16***b* side. The through-holes **16***a***1**, **16***a***2**, **16***a***3** are holes to cause a face of the bearing support part **16***a* on the motor **30** side and a face on the compressing part **20** side to communicate with each other.

As illustrated in FIG. **2**, grooves **16***a***4** communicating with the through-holes **16***a***1** and the second space IS**2** are formed on a face of the bearing support part **16***a* on the thrust plate **16***b* side. The grooves **16***a***4** are formed at two portions so as to extend in the radial direction orthogonal to the axis X. The grooves **16***a***4** guide the mixed refrigerant guided from the through-holes **16***a***1** radially to the region in which the bearing part **50** of the second space IS**2** is arranged. In FIG. **2**, the grooves **16***a***4** are illustrated with hatching. The same applies to a groove **16***a***5** described later. Further, in FIG. **4** and FIG. **6**, the groove **16***a***5** is also illustrated with hatching.

As illustrated in FIG. **2**, the groove **16***a***5** communicating with the second space IS**2** is formed on a face of the bearing support part **16***a* on the thrust plate **16***b* side. The groove **16***a***5** is formed at one portion at the upper end in the vertical direction VD of the bearing support part **16***a* so as to extend in the radial direction orthogonal to the axis X. The width of the inlet region (inner circumference side region) of the groove **16***a***5** in the rotation direction RD (the circumferential direction about the axis X) is W**1**. The width of the outlet region (outer circumference side region) of the groove **16***a***5** in the rotation direction RD is W**2** that is larger than W**1**.

The thrust plate **16***b* is a plate-like member installed to the bearing support part **16***a* and supporting the orbiting scroll **22** by a sliding surface **16***b*A that is in contact with an end face **22***a* of the orbiting scroll **22** of the compressing part **20**. FIG. **3** is a plan view of the thrust plate **16***b* illustrated in FIG. **1**. As illustrated in FIG. **3**, a notch **16***b***1** arranged at a position corresponding to the groove **16***a***5** formed in the bearing support part **16***a* is formed in the thrust plate **16***b*. The notch **16***b***1** has a shape that blocks the inlet region of the groove **16***a***5** while avoiding and thus not blocking the outlet region of the groove **16***a***5**.

The width of the notch **16***b***1** in the rotation direction RD is W**3**. In the rotation direction RD, the width W**1** of the inlet region of the groove **16***a***5** of the bearing support part **16***a* is smaller than the width W**3** of the notch **16***b***1**. The width W**3** of the notch **16***b***1** substantially matches the width W**2** of the outlet region of the groove **16***a***5** of the bearing support part **16***a*.

In the thrust plate **16***b*, notches **16***b***2** arranged at positions corresponding to the through-holes **16***a***2** formed in the bearing support part **16***a* are formed. The notches **16***b***2** have a shape that avoids and thus does not block the region overlapping the through-holes **16***a***2**. In the rotation direction RD, the width of the notch **16***b***2** substantially matches the width of the through-hole **16***a***2** of the bearing support part **16***a*.

In the thrust plate **16***b*, notches **16***b***3** arranged at positions corresponding to the through-holes **16***a***3** formed in the bearing support part **16***a* are formed. The notches **16***b***3** have a shape that avoids and thus does not block the region overlapping the through-holes **16***a***3**. In the rotation direction

RD, the width of the notch **16***b***3** substantially matches the width of the through-hole **16***a***3** of the bearing support part **16***a*.

FIG. **4** is a diagram illustrating a state where the thrust plate **16***b* is installed to the housing **10** illustrated in FIG. **2**. As illustrated in FIG. **4**, the through-holes **16***a***1**, the grooves **16***a***4**, and the groove **16***a***5** formed in the bearing support part **16***a* are covered with a facing surface **16***b*B of the thrust plate **16***b* (see FIG. **1**). The facing surface **16***b*B is a face that faces the face of the bearing support part **16***a* on the thrust plate **16***b* side.

The mixed refrigerant guided to the through-holes **16***a***1** from the first space IS**1** collides with the facing surface **16***b*B of the thrust plate **16***b* and then is guided to the grooves **16***a***4** communicating with the through-holes **16***a***1**. The mixed refrigerant guided to the grooves **16***a***4** is guided toward the axis X in the radial direction orthogonal to the axis X. The region to which the mixed refrigerant is guided is a region in which the bearing part **50** of the second space IS**2** is arranged.

As discussed above, the partition wall part **16** formed of the bearing support part **16***a* and the thrust plate **16***b* has first communication passages **16***c* for guiding the mixed refrigerant containing a lubricating oil and a refrigerant gas from the first space IS**1** to a region in which the bearing part **50** of the second space IS**2** is arranged (see FIG. **1**). The first communication passages **16***c* are formed of the through-holes **16***a***1**, the grooves **16***a***4**, and the facing surface **16***b*B of the thrust plate **16***b*.

The mixed refrigerant guided to the region in which the bearing part **50** of the second space IS**2** is arranged flows through in the rotation direction RD, flows into the inlet region of the groove **16***a***5** of the bearing support part **16***a*, and is guided to the outlet region. The refrigerant gas guided to the outlet region of the groove **16***a***5** is guided via the notch **16***b***1** to the region in which the compressing part **20** of the second space IS**2** is arranged.

As discussed above, the partition wall part **16** has a second communication passage **16***d* for guiding, to the compressing part **20**, a refrigerant gas contained in the mixed refrigerant guided to a region in which the bearing part **50** of the second space IS**2** is arranged (see FIG. **1**). The second communication passage **16***d* is formed of the groove **16***a***5**, the facing surface **16***b*B of the thrust plate **16***b*, and the notch **16***b***1**.

The mixed refrigerant guided from the first space IS**1** to the through-holes **16***a***2** is guided from the first space IS**1** to intake positions P**3** of the compressing part **20** of the second space IS**2** without passing through the region in which the bearing part **50** is arranged. This is because the notches **16***b***2** are formed at positions corresponding to the through-holes **16***a***2** of the thrust plate **16***b*. The intake position P**3** is an end (winding end) of the fixed scroll **21**. Further, the intake position P**3** is an end (winding end) of the orbiting scroll **22**.

As discussed above, the partition wall part **16** has third communication passages **16***e* configured to guide the refrigerant gas from the first space IS**1** to the compressing part **20** of the second space IS**2** without causing the refrigerant gas to pass through the region in which the bearing part **50** is arranged. The third communication passage **16***e* guides the refrigerant gas to the intake position P**3** in which the end of the fixed scroll **21** or the end of the orbiting scroll **22** is arranged.

The mixed refrigerant guided from the first space IS**1** to the through-holes **16***a***3** is guided from the first space IS**1** to the compressing part **20** of the second space IS**2** without passing through the region in which the bearing part **50** is arranged. This is because the notches **16***b***3** are formed at positions corresponding to the through-holes **16***a***3** of the thrust plate **16***b*.

A notch (oil returning part) **16***b***5** for returning the lubricating oil (not illustrated) that remains below the housing **10** in the vertical direction VD from the sliding surface **16***b*A side to the facing surface **16***b*B side is formed in the thrust plate **16***b*. As illustrated in FIG. **4**, the second communication passage **16***d* is formed on the opposite side of the notch **16***b***5** with respect to the axis X. The second communication passage **16***d* is formed at a position away from the notch **16***b***5** in the circumferential direction about the axis X in an angular range of 90 degrees or more and 270 degrees or less. By forming the second communication passage **16***d* on the opposite side of the notch **16***b***5** with respect to the axis X, it is possible to suppress the lubricating oil from being guided to the second communication passage **16***d*.

In the example illustrated in FIG. **4**, the first communication passages **16***c* are arranged at two portions, one of which is at about 135 degrees away from the position at which the second communication passage **16***d* is arranged in the opposite direction of the rotation direction RD, and the other of which is at about 315 degrees away from the position at which the second communication passage **16***d* is arranged. In such a case, in the rotation direction RD, the positions at which the first communication passages **16***c* are arranged and the position at which the second communication passage **16***d* is arranged are away from each other by about 135 degrees or more.

The example illustrated in FIG. **4** may be changed to other modified examples. For example, in the rotation direction RD, the position at which the first communication passage **16***c* is arranged and the position at which the second communication passage **16***d* is arranged may be away from each other by 180 degrees or more. FIG. **6** is a diagram of the housing **10** of a scroll compressor **100**A in the modified example when viewed from the compressing part **20** side. In FIG. **6**, the first communication passage **16***c* is arranged at only one portion, which is at about 315 degrees away from the position at which the second communication passage **16***d* is arranged in the opposite direction of the rotation direction RD. In such a case, in the rotation direction RD, the position at which the first communication passage **16***c* is arranged and the position at which the second communication passage **16***d* is arranged are away from each other by about 315 degrees.

In the rotation direction RD, when the position at which the first communication passage **16***c* is arranged and the position at which the second communication passage **16***d* is arranged are more away from each other, the lubricating oil contained in the refrigerant mixture guided from the first communication passage **16***c* to the region in which the bearing part **50** of the second space IS**2** is arranged is more suppressed from being guided to the second communication passage **16***d*. This is because the lubricating oil is more likely to be separated from the refrigerant gas for a larger length in the rotation direction RD from the first communication passage **16***c* to the second communication passage **16***d*.

In the example illustrated in FIG. **6**, although the first communication passage **16***c* is arranged at only one portion, which is at about 315 degrees away from a position at which the second communication passage **16***d* is arranged in the opposite direction of the rotation direction RD, various modifications are possible. For example, in the rotation direction RD, another position may be used as long as the position at which the first communication passage **16***c* is arranged and the position at which the second communication passage **16***d* is arranged are away from each other by 180 degrees or more.

The advantageous effects achieved by the scroll compressor **100** of the present embodiment described above will be described.

According to the scroll compressor **100** of the present embodiment, the rotary shaft **40** is rotated about the axis X by the motor **30**, the orbiting scroll **22** to which the rotary shaft **40** is installed via the eccentric shaft **41** is revolved with respect to the fixed scroll **21**, and the refrigerant gas is compressed and discharged. The inner space IS of the housing **10** for accommodating the motor **30** and the compressing part **20** is partitioned by the partition wall part **16** into the first space IS**1** in which the motor **30** is arranged and the second space IS**2** in which the bearing part **50** and the compressing part **20** are arranged.

The mixed refrigerant containing the lubricating oil and the refrigerant gas is guided by the first communication passage **16***c* in the partition wall part **16** from the first space IS**1** to the region in which the bearing part **50** of the second space IS**2** is arranged. The refrigerant gas contained in the mixed refrigerant guided to the region in which the bearing part **50** of the second space IS**2** is arranged is guided by the second communication passage **16***d* to the compressing part **20**. In such a way, according to the scroll compressor **100** of the present embodiment, it is possible to reliably guide the lubricating oil contained in a mixed refrigerant to the bearing part **50** and reliably guide the refrigerant gas contained in the mixed refrigerant to the compressing part **20**.

According to the scroll compressor **100** of the present embodiment, the mixed refrigerant supplied to the first space IS**1** from the intake port P**1** can be passed through the through-holes **16***a***1**, guided to the grooves **16***a***4**, passed between the grooves **16***a***4** and the facing surface **16***b*B of the thrust plate **16***b*, and guided to the region in which the bearing part **50** of the second space IS**2** is arranged.

According to the scroll compressor **100** of the present embodiment, the refrigerant gas contained in the mixed refrigerant supplied to the second space IS**2** can be passed between the groove **16***a***5** and the facing surface **16***b*B of the thrust plate **16***b* and guided from the notch **16***b***1** to the compressing part **20**.

According to the scroll compressor **100** of the present embodiment, the refrigerant gas is guided by the third communication passage **16***e* to the intake position P**3** of the compressing part **20** of the second space IS**2** from the first space IS**1** without passing through the region in which the bearing part **50** is arranged. It is thus possible to directly guide the refrigerant gas to the intake position P**3** of the compressing part **20** without causing a heat loss due to the refrigerant gas passing through the bearing part **50** and being heated.

According to the scroll compressor **100** of the present embodiment, the second communication passage **16***d* is formed on the opposite side to a position at which the notch **16***b***5** is formed with respect to the axis X. It is thus possible to suitably prevent the lubricating oil from being guided to the second communication passage **16***d*.

According to the scroll compressor **100** of the present embodiment, since the width W**1** of the groove **16***a***5** is smaller than the width W**3** of the notch **16***b***1**, it is possible to suppress the lubricating oil contained in the mixed refrigerant from being guided from the notch **16***b***1** to the compressing part **20** compared to the case where the width of the groove **16***a***5** is the same as the width of the notch **16***b***1**.

According to the scroll compressor **100** of the present embodiment, when the position at which the first communication passage **16***c* is arranged and the position at which the second communication passage **16***d* is arranged are away from each other by 180 degrees or more, the lubricating oil that has flown into the second space IS**2** from the first communication passage **16***c* can be moved in the rotation direction RD by 180 degrees or more. Thus, compared to the case where the angle by which the position at which the first communication passage **16***c* is arranged and the position at which the second communication passage **16***d* is arranged are away from each other is less than 180 degrees in the rotation direction, the amount of lubricating oil that remains around the bearing part **50** can be increased.

The scroll compressor according to the present embodiment described above is understood as follows, for example.

A scroll compressor (**100**) according to the present disclosure includes: a compressing part (**20**) having a fixed scroll (**21**) and an orbiting scroll (**22**) engaged with the fixed scroll; a motor (**30**) configured to revolve the orbiting scroll with respect to the fixed scroll; a rotary shaft (**40**) configured to be rotated about an axis (X) by the motor and installed to the orbiting scroll via an eccentric shaft (**41**) arranged eccentrically from the axis; a bearing part (**50**) configured to support the rotary shaft; and a housing (**10**) formed in a cylindrical shape along the axis and having an inner space (IS) for accommodating the motor and the compressing part. The housing has a partition wall part (**16**) partitioning the inner space (IS) into a first space (IS**1**) and a second space (IS**2**), the motor being arranged in the first space, and the bearing part and the compressing part being arranged in the second space. The partition wall part has a first communication passage (**16***c*) and a second communication passage (**16***d*), the first communication passage being configured to guide a mixed refrigerant containing a lubricating oil and a refrigerant gas from the first space to a region of the second space, the bearing part being arranged in the region of the second space, and the second communication passage being configured to guide the refrigerant gas, which is contained in the mixed refrigerant guided to the region of the second space, to the compressing part.

According to the scroll compressor of the present disclosure, the rotary shaft is rotated about the axis by the motor, the orbiting scroll to which the rotary shaft is installed via the eccentric shaft is revolved with respect to the fixed scroll, and the refrigerant gas is compressed and discharged. The inner space of the housing for accommodating the motor and the compressing part is partitioned by the partition wall part into the first space in which the motor is arranged and the second space in which the bearing part and the compressing part is arranged.

The mixed refrigerant containing the lubricating oil and the refrigerant gas is guided by the first communication passage in the partition wall part from the first space to the region in which the bearing part of the second space is arranged. The refrigerant gas contained in the mixed refrigerant guided to the region in which the bearing part of the second space arranged is guided by the second communication passage to the compressing part. In such a way, according to the scroll compressor of the present disclosure, it is possible to reliably guide the lubricating oil contained in the mixed refrigerant to the bearing part and reliably guide the refrigerant gas contained in the mixed refrigerant to the compressing part.

The scroll compressor according to the present disclosure may be configured such that the partition wall part has a bearing support part (**16***a*) partitioning the inner space into the first space and the second space, the bearing part being press-fitted into the bearing support part, and a plate-like thrust plate (**16***b*) installed to the bearing support part and configured to support the orbiting scroll by a sliding surface (**16***b*A) that is in contact with an end face of the orbiting scroll, and the first communication passage (**16***c*) is formed of a through-hole (**16***a***1**) formed in the bearing support part, a first groove (**16***a***4**) formed in a face of the bearing support part on the thrust plate side and communicating with the through-hole and the second space, and a facing surface (**16***b*B) of the thrust plate on the bearing support part side, the thrust plate being arranged so as to cover the first groove.

According to the scroll compressor of the present configuration, the mixed refrigerant supplied to the first space can be passed through the through-holes, guided to the first groove, passed between the first groove and the facing surface of the thrust plate, and guided to the region in which the bearing part of the second space is arranged.

The scroll compressor according to the present disclosure may be configured such that the second communication passage (**16***d*) is formed of a second groove (**16***a***5**) formed in a face of the bearing support part on the thrust plate side, a facing surface of the thrust plate on the bearing support part side, the thrust plate being arranged so as to cover the second groove, and a notch formed in the thrust plate and configured to guide the refrigerant gas from the second groove to the compressing part.

According to the scroll compressor of the present configuration, the refrigerant gas contained in the mixed refrigerant supplied to the second space can be passed between the second groove and the facing surface of the thrust plate and guided from the notch to the compressing part.

The scroll compressor according to the present disclosure may be configured such that the partition wall part has a bearing support part partitioning the inner space into the first space and the second space, the bearing part being press-fitted into the bearing support part, and a plate-like thrust plate installed to the bearing support part and configured to support the orbiting scroll by a sliding surface that is in contact with an end face of the orbiting scroll, and the second communication passage is formed of a second groove formed in a face of the bearing support part on the thrust plate side, a facing surface on the bearing support part side of the thrust plate arranged so as to cover the second groove, and a notch formed in the thrust plate and configured to guide the refrigerant gas from the second groove to the compressing part.

According to the scroll compressor of the present configuration, the refrigerant gas contained in the mixed refrigerant supplied to the second space can be passed between the second groove and the facing surface of the thrust plate and guided from the notch to the compressing part.

The scroll compressor according to the present disclosure may be configured such that the partition wall part has a third communication passage (**16***e*) configured to guide the refrigerant gas from the first space to the compressing part of the second space without causing the refrigerant gas to pass through the region in which the bearing part is arranged, and the third communication passage guides the refrigerant gas to an intake position (P**3**) in which an end of the fixed scroll or an end of the orbiting scroll is arranged.

According to the scroll compressor of the present configuration, the refrigerant gas is guided by the third communication passage to the intake position of the compressing part of the second space from the first space without passing through the region in which the bearing part is arranged. It is thus possible to directly guide the refrigerant gas to the intake position of the compressing part without causing a heat loss due to the refrigerant gas passing through the bearing part and being heated.

The scroll compressor according to the present disclosure may be configured such that, in the thrust plate, an oil returning part (**16***b***5**) configured to return the lubricating oil that remains below the housing from the sliding surface side to the facing surface side is formed, and the second communication passage is formed on the opposite side, with respect to the axis, to a position at which the oil returning part is formed.

According to the scroll compressor of the present configuration, the second communication passage is formed on the opposite side to a position at which the oil returning part is formed with respect to the axis. It is thus possible to suitably prevent the lubricating oil from being guided to the second communication passage.

The scroll compressor according to the present disclosure may be configured such that, in a rotation direction of the rotary shaft, the width of the second groove is smaller than the width of the notch.

According to the scroll compressor of the present configuration, since the width of the second groove is smaller than the width of the notch, it is possible to suppress the lubricating oil contained in the mixed refrigerant from being guided from the notch to the compressing part compared to the case where the width of the second groove is the same as the width of the notch.

The scroll compressor according to the present disclosure may be configured such that, in a rotation direction of the rotary shaft, a position at which the first communication passage is arranged and a position at which the second communication passage is arranged are away from each other by 180 degrees or more.

According to the scroll compressor of the present configuration, since the position at which the first communication passage is arranged and the position at which the second communication passage is arranged are away from each other by 180 degrees or more, the lubricating oil that has flown into the second space from the first communication passage can be moved in the rotation direction by 180 degrees or more. Thus, compared to the case where the angle by which the position at which the first communication passage is arranged and the position at which the second communication passage is arranged is away from each other is less than 180 degrees in the rotation direction, the amount of lubricating oil that remains around the bearing part can be increased.

REFERENCE SIGNS LIST

10 housing
11 first housing
12 second housing
13 third housing
14 bolt
15 leg
16 partition wall part
16*a* bearing support part
16*a***1**, **16***a***2**, **16***a***3** through-hole
16*a***4**, **16***a***5** groove
16*b* thrust plate
16*b***1**, **16***b***2**, **16***b***3**, **16***b***5** notch
16*b*A sliding surface
16*b*B facing surface
16*c* first communication passage
16*d* second communication passage

16*e* third communication passage
20 compressing part
21 fixed scroll
22 orbiting scroll
22*a* end face
22*b* bearing part
30 motor
40 rotary shaft
41 eccentric shaft
50, **60** bearing part
70 balance weight
80 inverter
100, **100**A scroll compressor
IS inner space
IS**1** first space
IS**2** second space
P**1** intake port
P**3** intake position
RD rotation direction
VD vertical direction
X axis

The invention claimed is:

1. A scroll compressor comprising:
a compressing part having a fixed scroll and an orbiting scroll engaged with the fixed scroll;
a motor configured to revolve the orbiting scroll with respect to the fixed scroll;
a rotary shaft configured to be rotated about an axis by the motor and installed to the orbiting scroll via an eccentric shaft arranged eccentrically from the axis;
a bearing part configured to support the rotary shaft; and
a housing formed in a cylindrical shape along the axis and having an inner space for accommodating the motor and the compressing part,
wherein the housing has a partition wall part partitioning the inner space into a first space and a second space, the motor being arranged in the first space, and the bearing part and the compressing part being arranged in the second space, and
wherein the partition wall part has a first communication passage and a second communication passage, the first communication passage being configured to guide a mixed refrigerant containing a lubricating oil and a refrigerant gas from the first space to a region of the second space, the bearing part being arranged in the region of the second space, and the second communication passage being configured to guide the refrigerant gas, which is contained in the mixed refrigerant guided to the region of the second space, to the compressing part.

2. The scroll compressor according to claim **1**,
wherein the partition wall part has
a bearing support part partitioning the inner space into the first space and the second space, the bearing part being press-fitted into the bearing support part, and
a plate-like thrust plate installed to the bearing support part and configured to support the orbiting scroll by a sliding surface that is in contact with an end face of the orbiting scroll, and
wherein the first communication passage is formed of a through-hole formed in the bearing support part, a first groove formed in a face of the bearing support part on the thrust plate side and communicating with the through-hole and the second space, and a facing surface of the thrust plate on the bearing support part side, the thrust plate being arranged so as to cover the first groove.

3. The scroll compressor according to claim **2**, wherein the second communication passage is formed of a second groove formed in a face of the bearing support part on the thrust plate side, a facing surface of the thrust plate on the bearing support part side, the thrust plate being arranged so as to cover the second groove, and a notch formed in the thrust plate and configured to guide the refrigerant gas from the second groove to the compressing part.

4. The scroll compressor according to claim **3**,
wherein, in the thrust plate, an oil returning part configured to return the lubricating oil that remains below the housing from a sliding surface side to a facing surface side is formed, and
wherein the second communication passage is formed on the opposite side, with respect to the axis, to a position at which the oil returning part is formed.

5. The scroll compressor according to claim **3**, wherein in a rotation direction of the rotary shaft, a width of an inlet region of the second groove is smaller than a width of the notch.

6. The scroll compressor according to claim **1**,
wherein the partition wall part has
a bearing support part partitioning the inner space into the first space and the second space, the bearing part being press-fitted into the bearing support part, and
a plate-like thrust plate installed to the bearing support part and configured to support the orbiting scroll by a sliding surface that is in contact with an end face of the orbiting scroll, and
wherein the second communication passage is formed of a second groove formed in a face of the bearing support part on the thrust plate side, a facing surface of the thrust plate on the bearing support part side, the thrust plate being arranged so as to cover the second groove, and a notch formed in the thrust plate and configured to guide the refrigerant gas from the second groove to the compressing part.

7. The scroll compressor according to claim **1**,
wherein the partition wall part has a third communication passage configured to guide the refrigerant gas from the first space to the compressing part of the second space without causing the refrigerant gas to pass through the region in which the bearing part is arranged, and
wherein the third communication passage guides the refrigerant gas to an intake position in which an end of the fixed scroll or an end of the orbiting scroll is arranged.

8. The scroll compressor according to claim **1**, wherein in a rotation direction of the rotary shaft, a position at which the first communication passage is arranged and a position at which the second communication passage is arranged are away from each other by 180 degrees or more.

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